Polytope Notion	**Definition on P**	**Divisor/Data on X_P**	**Toric Variety Property**
Divisor–Polytope Correspondence	$P_D = \{m \in M_{\mathbb{R}} : \langle m, u_\rho \rangle \geq -a_\rho\}$.	$D_P = \sum_\rho a_\rho D_\rho$.	$X_P \cong Proj\left(\bigoplus_{m \geq 0} H^0(X, \mathcal{O}(mD_P))\right)$.
Lattice Polytope & Sections	$P \cap M = \{m_1, \ldots, m_s\}$.	$H^0(X_P, \mathcal{O}(D_P)) = Span\{\chi^{m_i}\}$.	$\phi_{D_P} : X_P \to \mathbb{P}^{s-1}$ via χ^{m_i}.
Very Ample Polytope	$\mathbb{N}(P \cap M - m_i)$ saturated at each vertex m_i.	D_P very ample.	ϕ_{D_P} is a closed embedding.
Ample Polytope	$\psi_D : \lvert\Sigma\rvert \to \mathbb{R}$ strictly convex.	$\lvert D_P\rvert$ ample.	ϕ_{mD_P} is a closed embedding for some m.
Base-point-Free (or Nef??) Polytope	$\psi_D : \lvert\Sigma\rvert \to \mathbb{R}$ convex.	$\lvert D_P\rvert$ has no base-points.	ϕ_{D_P} defined everywhere.
Normal Polytope	$(kP \cap M) + (lP \cap M) = ((k+l)P \cap M)$ for all k, l.	Section ring $\bigoplus_{m \geq 0} H^0(\mathcal{O}(mD_P))$ is integrally closed.	Embedding is projectively normal.
Reflexive Polytope	$0 \in Int(P)$ and dual P° is lattice.	$D_{P^\circ} = -K_{X_P}$.	X_P is Gorenstein Fano.
Smooth Polytope	At each vertex, the primitive edge-directions form a $\mathbb{Z}$-basis of M.	Equivalently, all cones in the normal fan are simplicial and unimodular.	X_P is a smooth toric variety (no singularities).
Toric Resolution	A regular (unimodular) refinement Σ' of the normal fan Σ_P.	Induces $\pi\colon X_{\Sigma'} \to X_{\Sigma_P}$, a proper birational toric morphism.	$X_{\Sigma'}$ is smooth and π resolves the singularities of X_P.
Minkowski Sum	$P + Q = \{p + q \mid p \in P, q \in Q\}$.	$D_{P+Q} = D_P + D_Q$.	$H^0(\mathcal{O}(D_{P+Q})) = H^0(\mathcal{O}(D_P)) \cdot H^0(\mathcal{O}(D_Q))$.
Ehrhart = Hilbert	$E_P(m) = \lvert mP \cap M\rvert$.	$\dim H^0(\mathcal{O}(mD_P)) = E_P(m)$.	Hilbert polynomial of $\mathcal{O}(D_P)$ is $E_P(m)$.

Table 1: Correspondence between polytope properties and the associated toric divisor on X_P.